

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of MSc Microbiology Examination April/May 2018

Course code & Name MI 761 & Microbial Genetics

Date: 5/5/2018 Day: Saturday Time: 1:30 pm To 2:00 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	Mutation in the tumor suppressor gene is referred to as ____		01
	(i) Silent mutation	(ii) Neutral mutation	
	(iii) Gain of function mutation	(iv) Loss of function mutation	
2.	____ acts as anti-terminator for transcription of late genes.		01
	(i) N	(ii) Cro	
	(iii) Q	(iv) C-II	
3.	____ is a temperate phage, whose DNA does not integrate within the host bacterial genome.		01
	(i) P1	(ii) λ	
	(iii) T4	(iv) None of the above	
4.	The unit for gene mapping during interrupted mating experiment is ____		01
	(i) bp	(ii) kD	
	(iii) Time	(iv) All of the above	

5.	The first transposable elements found in the prokaryotic cells were ____		01
	(i) Transposons	(ii) P elements	
	(iii) IS elements	(iv) Mu	
6.	Removal of plasmid from the progeny due to unequal distribution, is called as ____		01
	(i) Plasmid curing	(ii) Plasmid incompatibility	
	(iii) Plasmid infertility	(iv) None of the above	
7.	Mutation in the DNA binding domain of <i>Lac I</i> protein leads to ____		01
	(i) Operon remains switched off	(ii) Operon remains switched on	
	(iii) ATP level will increase	(iv) C-AMP level will increase	
8.	Conjugation between F^+ and F^- cells always leads to generation ____ cells.		01
	(i) F^+	(ii) F^-	
	(iii) Hfr	(iv) F'	
9.	Purine is replaced by Purine due to mutation. It is called as ____.		01
	(i) Transition	(ii) Transversion	
	(iii) Transposition	(iv) Transfection	
10.	Mode of action of EMS is ____.		01
	(i) Dimer formation	(ii) Intercalation	
	(iii) Alkylation	(iv) Oxidation	
11.	____ is a virulent phage, which does not lyse the host cells during release.		01
	(i) P1	(ii) fd	
	(iii) T4	(iv) λ	
12.	Production of tetrads classified as ____ during meiosis in a dihybrid, due to independent assortment of two genes located on different chromosomes.		01
	(i) Non-parental ditype (NPD)	(ii) Parental ditype (PD)	
	(iii) Tetratype tetrads	(iv) None of the above	
13.	Which of the following is the most appropriate definition of an operator?		01
	(i) A non-coding, regulatory DNA sequence that is bound by RNA polymerase	(ii) A non-coding, regulatory DNA sequence that is bound by a repressor protein	
	(iii) A DNA-binding protein that regulates gene expression	(iv) A cluster of genes that are regulated by a single promoter	
14.	Transfer of a gene in animal cell is called as ____.		01
	(i) Transformation	(ii) Conjugation	
	(iii) Both, (i) and (ii)	(iv) Transfection	

15.	_____ resolves the holiday junction model.		01
	(i) Ruv A	(ii) Ruv B	
	(iii) Ruv C	(iv) All of the above	
16.	Ti-plasmid is an example of _____ type of plasmids.		01
	(i) Conjugative plasmid	(ii) Virulent	
	(iii) Col	(iv) Both, (i) and (ii)	
17.	_____ is not a structural gene in the Lac operon.		01
	(i) Lac I	(ii) Lac Z	
	(iii) Lac Y	(iv) Lac A	
18.	_____ introduces gene directly in the nucleus of an animal cell.		01
	(i) Microinjection	(ii) Protoplast fusion	
	(iii) Cell squeezing	(iv) Use of dendrimers	
19.	Among the following, statement _____ is incorrect in reference to natural transformation.		01
	(i) Competence factor is secreted out from the Gram positive bacteria	(ii) Competence factor remains in the periplasmic space of the Gram negative bacteria	
	(iii) Only homologous DNA can bind get transformed efficiently in Gram positive cells.	(iv) Only homologous DNA can bind get transformed efficiently in Gram negative cells.	
20.	_____ leads to numerical abnormality.		01
	(i) Deletion	(ii) Inversion	
	(iii) Translocation	(iv) Nondisjunction	

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II Semester of MSc Microbiology Examination April/May 2018

Course code & Name MI 761 & Microbial Genetics

Date: 5/5/2018 Day: Saturday Time: 2:00 pm To 4:30 pm

Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEETS.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I										
Q - II Answer the following questions as directed (Any 10)		20								
1	What is sexduction?	02								
2	What is the principle of pulse field electrophoresis?	02								
3	Differentiate: Silent, Missense and Nonsense mutation	02								
4	Define: ordered and unordered tetrads	02								
5	What is specialized transduction?	02								
6	Differentiate: composite and non-composite transposons	02								
7	What is immunity to superinfection during bacteriophage infection?	02								
8	Define: Mutation rate and Mutation frequency	02								
9	The Hfr strain possessing the markers $his^+ met^+ tyr^+ str^s$ was mated with a F^- strain possessing the markers $his^- met^- tyr^- str^r$. The time of entry for each marker is shown below. <table><tr><td><u>Donor Marker</u></td><td><u>Time (min)</u></td></tr><tr><td>his^+</td><td>21</td></tr><tr><td>met^+</td><td>12</td></tr><tr><td>tyr^+</td><td>46</td></tr></table> Based on the above results, • Draw a map of the Hfr chromosome. • Indicate the position of the F element, and the first marker to be transferred in the same.	<u>Donor Marker</u>	<u>Time (min)</u>	his^+	21	met^+	12	tyr^+	46	02
<u>Donor Marker</u>	<u>Time (min)</u>									
his^+	21									
met^+	12									
tyr^+	46									
10	What is Phenotypic lag?	02								
11	Explain roles of proteins involved in the regulation of recombination.	02								
12	What is one step growth curve?	02								

SECTION – II		
Q-III	Answer the following questions as directed (Any 6)	30
1	Explain the statement: ‘The lac operon is positively and Negatively regulated.’ (with suitable labeled figures)	05
2	Explain the mechanisms behind regulation of the trp operon (with suitable figures)	05
3	Write a short note on Ti-plasmid and roles of vir genes in the process of infection (with suitable figure)	05
4	Why the term transfection is used rather than transformation? Explain chemical methods used in transfection.	05
5	Explicate in detail: principle, procedure, applications and advancements in the Ames test	05
6	Elucidate roles of Cro, N and Q during infection by a temperate phage (with suitable labelled figures).	05
7	What is Phage therapy? (with suitable figures)	05